

MICHAEL GIZAW

michaelhgizaw@gmail.com • 202-509-4020 • michaelgizaw.com

EDUCATION

University of Maryland, College Park *Expected August 2026*

B.S. in Computer Science · College Park, MD

- Member, **Smith Analytics Consortium** — an industry–academia analytics partnership (Deloitte, KPMG, Unilever).
- **Relevant Coursework:** Algorithms, Data Structures, Computer Systems, Computer & Network Security (CMSC414), Organization of Programming Languages, Data Science, Web Application Development.

TECHNICAL SKILLS

Languages: Python, C, Rust, JavaScript, SQL, OCaml, HTML/CSS

Frameworks & Libraries: React, NumPy, Pandas, Pygame

Tools & Platforms: Docker, Git, Linux, Google Colab, Google Earth Engine, NoSQL, Dafny

Security: Buffer overflow exploitation, SQL injection, XSS, CSRF, applied cryptography

Spoken Languages: English, Amharic

PROJECTS

Wildfire Burn-Severity Mapping *Python · Google Earth Engine · Random Forest*

- Built an end-to-end Google Earth Engine pipeline to map wildfire burn severity across Yengo National Park, Australia (2019–20 bushfire season), fusing Landsat 8 and Sentinel-2 imagery.
- Computed differenced Normalized Burn Ratio (dNBR) from pre/post-fire composites and classified terrain into six USGS-style severity levels using a 300-tree Random Forest (`smileRandomForest`) over a 10-band feature stack.
- Trained on stratified random samples (70/30 split) and validated with a confusion matrix, overall/producer/consumer accuracy, and feature-importance analysis.

News Article Text Classification *Python · NumPy*

- Built models to automatically categorize news articles by topic (e.g., sports → soccer), reaching 83% classification accuracy.
- Implemented a neural network and a Support Vector Machine (SVM) from scratch — no ML frameworks — and compared their accuracy and performance trade-offs.
- Owned the full ML workflow: text preprocessing, feature extraction, model training, and evaluation.

Facial Recognition Neural Network *Python · NumPy*

- Designed and implemented an artificial neural network from scratch to identify and classify individuals from facial image data.
- Engineered the data pipeline, including one-hot encoding of labels and image preprocessing for model input.
- Iterated on network architecture and training to improve recognition performance.

Web & Systems Security *C · Web Security · Cryptography*

- Exploited buffer overflow vulnerabilities to understand memory-safety attacks and the defenses that prevent them.
- Carried out SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF) attacks against test applications, then applied the corresponding mitigations.
- Worked with NoSQL data stores and applied cryptographic primitives to secure stored and transmitted data.

EXPERIENCE

AV Technician *June 2024 – Present*

University of Maryland · College Park, MD

- Provide on-site technical support for audiovisual systems across all general-purpose classrooms and most academic buildings campus-wide.
- Diagnose and resolve hardware and software issues in real time — including conferencing tools such as Zoom — to keep classes and live events running without disruption.
- Act as first responder for AV failures during lectures and events, guiding non-technical faculty through fixes clearly and quickly.

Office Assistant *June 2023 – August 2023*

University of Maryland · College Park, MD

- Coordinated a department-wide move-out and move-in during facility renovations, liaising across multiple teams to ensure a smooth transition.
- Managed asset inventory end-to-end, ensuring safe transport and organized resettlement of equipment into refurbished spaces.

Summer Camp Instructor & Mentor *May 2022 – August 2022*

Drobots Company · Washington, DC

- Taught coding with Scratch to students aged 6–14, led daily camp operations, and troubleshooted drone hardware.